

Machine Learning 2 - Assignment #2

Data Science Specialization

University of Antioquia

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Overview

In this assignment, you will explore **dimensionality reduction techniques** and their impact on clustering tasks. You will apply **PCA, Kernel PCA, t-SNE, and UMAP** to project high-dimensional data into a lower-dimensional space and analyze whether these techniques improve the separability of clusters.

Furthermore, you will evaluate the effectiveness of these embeddings by performing **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** on the transformed datasets.

Objectives

By the end of this assignment, students will be able to:

1. Implement and compare different dimensionality reduction techniques.
2. Analyze the quality of embeddings and their suitability for clustering.
3. Apply DBSCAN to assess the separability of clusters in reduced feature spaces.

Datasets

Students can choose one of the following datasets:

- **FIFA 23 Players Dataset:** Contains detailed attributes of professional soccer players. The goal is to investigate whether dimensionality reduction can enhance player segmentation based on skill sets.
 - Features of interest: Age, Overall rating, Potential, Value, Wage, Shooting, Passing, Dribbling, Defending, and Physicality.

Dataset available on Kaggle: [FIFA 23 Dataset](#) - Kaggle.

- **EastWestAirlines Dataset:** Contains airline customer behavior data. The objective is to assess whether dimensionality reduction aids in segmenting customer profiles based on usage patterns. Dataset available here: [EastWestAirlines Dataset](#).

Tasks

1. **Dimensionality Reduction:** Implement the following techniques and visualize their embeddings:

- **Principal Component Analysis (PCA)**
- **Kernel PCA**
- **t-SNE (t-Distributed Stochastic Neighbor Embedding)**
- **UMAP (Uniform Manifold Approximation and Projection)**

For each technique, analyze:

- How well the data is preserved in lower dimensions.
- Whether clusters appear more defined.

2. **Clustering with DBSCAN:** Apply **DBSCAN** to the reduced-dimensional data and evaluate:

- How clustering performance is affected by different embeddings.
- The number of clusters detected in each case.
- The presence of noise (points classified as outliers).

3. **Comparison and Analysis:**

- Compare the four dimensionality reduction methods in terms of cluster separability.
- Identify which method best preserves structure for clustering.
- Discuss whether the reduced feature space provides a meaningful representation for classification or decision-making.

Deliverables

- Teams must submit an **IPYNB notebook (Jupyter notebook)** containing:
 - Implementations of **PCA, Kernel PCA, t-SNE, and UMAP**.
 - Visual comparisons of embeddings (e.g., scatter plots).
 - Application of **DBSCAN** to analyze clustering behavior in the embedded space.
 - Interpretation and discussion of findings.
- The notebook should be well-documented with **Markdown cells** explaining each step and analysis.

Hints and Useful Links

- **PCA and Kernel PCA:** Scikit-learn: PCA and Kernel PCA.
- **t-SNE:** Scikit-learn: t-SNE.
- **UMAP:** UMAP Documentation.
- **DBSCAN:** Scikit-learn: DBSCAN.

Grading Rubric

- **Correctness of Implementation (40%):** Accurate implementation of dimensionality reduction techniques and clustering.
- **Evaluation and Analysis (30%):** Comparison of embeddings and their effects on clustering.
- **Visualization and Reporting (20%):** Well-structured figures and discussion.
- **Code Quality (10%):** Readability, documentation, and organization.

Submission

Please submit your Jupyter notebooks in IPYNB format, including all code, analysis, and documentation. The deadline for this assignment is **19/03/2025**.